MUHAMMAD FARID MASOOD KHAN

Applied Machine Learning & MLOps Engineer

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ABOUT ME & ENGINEERING PROFILE

Applied Machine Learning & MLOps Engineer with a Bachelor of Science in Computer Science (2024) and 1.5 years of professional software engineering experience. Specializes in designing robust, deterministic machine learning systems: point-in-time feature stores, continuous delivery pipelines (Feast, MLflow, DVC), low-latency quantized edge vision (ONNX Runtime), hybrid enterprise RAG with verifiable citations, and uncertainty-calibrated time series forecasting. Solid mathematical and statistical grounding across Stanford/DeepLearning.AI and Imperial College London specializations. Proven anti-'vibe coding' standard backed by **109 automated unit/integration tests (100% passing)** across 7 public open-source production repositories.

TECHNICAL SKILLS & COMPETENCIES

MLOps & Delivery:	Feast Feature Store, MLflow Model Registry, DVC Data Versioning, Docker, Great Expectations, CI/CD Actions, KS & PSI Drift Monitoring.
Machine Learning:	Scikit-Learn, LightGBM, XGBoost, PyTorch, Statsmodels, Prophet, PyTorch Geometric, PU-Learning, Calibration (ECE).
Computer Vision:	MobileNetV3, ONNX Runtime Quantization (INT8/FP16), OpenCV, Image Quality Triage (Laplacian Blur Gates).
NLP & Retrieval:	Sentence Transformers, BM25 Okapi, Cross-Encoder Reranking, Reciprocal Rank Fusion (RRF), Unicode Normalization Engines.
Backend & Serving:	FastAPI (Typed REST APIs), Pydantic v2, SQLModel, SQLite, Streamlit, Firebase Backend Services.
Engineering & Code:	Python 3.12 (Expert), Dart (Flutter), SQL, C++, Git, Strict Typing (Mypy), Code Quality (Ruff), Pytest.

FEATURED APPLIED ML & MLOPS SYSTEMS (ALL 100% TESTED & REPRODUCIBLE)

MLForge — Continuous Delivery MLOps Pipeline for Credit Risk

26/26 Tests Passing | 100% Green

Feast Feature Store · MLflow Registry · DVC · Great Expectations · LightGBM · FastAPI · Docker | github.com/Farid-Masood-Khan/mlforge-mlops

- Architected a production continuous delivery pipeline for credit risk scoring, eliminating train-serve skew via point-in-time correct Feast feature store joins.
- Enforced strict pre-training schema and value range gates using Great Expectations; integrated automated distribution drift monitoring via Kolmogorov-Smirnov (KS) tests and Population Stability Index (PSI).
- Managed automated Champion/Challenger model registry triage in MLflow with automated rollback safeguards; built high-throughput typed FastAPI serving microservice with full containerization.

CropGuard — Edge Agri-AI Vision & Calibrated Leaf Disease Diagnosis

17/17 Tests Passing | 100% Green

PyTorch · MobileNetV3-Small · ONNX Runtime · Platt Temperature Scaling · OpenCV · FastAPI | github.com/Farid-Masood-Khan/cropguard-agri-ai

- Engineered an edge computer vision classifier diagnosing 38 PlantVillage crop disease classes across 14 crops, optimized for rural smallholder agricultural deployment.
- Quantized PyTorch weights to a 16.1 MB ONNX Runtime engine achieving 2.1 ms CPU inference latency (14x faster than vanilla PyTorch on commodity hardware).
- Cut Expected Calibration Error (ECE) from 0.084 to 0.0257 using Platt temperature scaling; integrated a pre-inference Laplacian blur rejection gate and human agronomist deferral triage.

DocuMind — Enterprise Document RAG & Hybrid Retrieval Fusion

12/12 Tests Passing | 100% Green

FastAPI · Streamlit · BM25 Okapi · BAAI Cross-Encoder · Reciprocal Rank Fusion · SQLModel · SQLite | github.com/Farid-Masood-Khan/documind-rag

- Built an enterprise retrieval-augmented generation (RAG) chatbot delivering grounded document Q&A; with verifiable sentence-level citations to eliminate hallucinations.
- Designed a dual-channel retrieval architecture fusing dense semantic embeddings with BM25 Okapi lexical scores via Reciprocal Rank Fusion (RRF), followed by BAAI/bge-reranker cross-encoder precision re-ranking.
- Implemented sentence-window chunking preserving local surrounding context around matched tokens for coherent, evidence-backed generation.

TrendCast — Quantitative Food Commodity Time-Series Forecasting 18/18 Tests Passing | 100% Green

Prophet · SARIMAX · Statsmodels · Plotly · Streamlit · Rolling Backtesting | github.com/Farid-Masood-Khan/trendcast-forecasting

- Modeled 22 years of World Food Programme (WFP HDX) Pakistan food market retail price series (2004–2026) using additive Prophet and multiplicative SARIMAX models.
- Enforced leak-free expanding-window rolling-origin backtesting across 10 expanding folds, evaluated against seasonal persistence and naive baseline gates.
- Calibrated empirical prediction intervals on historical residuals achieving 81.8% coverage for honest uncertainty quantification.

UrduNLP Toolkit — Low-Resource NLP for Urdu & Roman Urdu 18/18 Tests Passing | 100% Green

Scikit-Learn · Python Regex Normalizer · Streamlit · Jaccard Splitting · Pytest | github.com/Farid-Masood-Khan/urdunlp-toolkit

- Engineered a specialized NLP normalizer for dual-script Urdu (Nastaliq Arabic script and informal Latin Roman Urdu), resolving high spelling entropy and character elongation.
- Achieved a measured 15.26% vocabulary reduction through Unicode diacritic stripping (Aerab), character unification rules, and subword n-gram extraction.
- Integrated Jaccard string similarity splitting to eliminate train-test leakage across conversational variants; calibrated tri-class sentiment classification.

Customer Churn Prediction & Retention Simulator 10/10 Tests Passing | 100% Green

Scikit-Learn · ColumnTransformer · Random Forest · Streamlit · Seaborn · Pytest | github.com/Farid-Masood-Khan/Customer-Churn-Prediction

- Developed an end-to-end leak-free tabular supervised learning pipeline on 7,043 IBM Telco customer accounts across 5-fold stratified cross-validation.
- Achieved 0.843 ROC-AUC and 79.4% recall (capturing 297/374 holdout churners); constructed an interactive Streamlit risk simulator; domain baseline for MLForge.

Sentiment Analysis of Product Reviews (Foundational Research Prototype) 8/8 Tests Passing | 100% Green

Scikit-Learn · Subword TF-IDF · Script Error Analysis · Regex Diacritic Stripper | github.com/Farid-Masood-Khan/Sentiment-Analysis-of-Product-Reviews

- Dual-script Urdu sentiment baseline analyzing orthographic variance between formal Arabic Nastaliq and Latin Roman Urdu transliteration; foundational precursor to UrduNLP.

Frugal — AI-Powered Food Waste & Redistribution Platform (Final Year Project) BSCS Thesis Project

Flutter · Firebase Firestore · Python ML Extension · Cloud Functions | github.com/Farid-Masood-Khan/Frugal

- Designed and shipped a full-stack cross-platform application connecting commercial food donors with NGOs; integrated demand forecasting regression and perishable shelf-life spoilage triage.

PROFESSIONAL EXPERIENCE

Flutter App Developer — Desired Technologies 11/2023 – 02/2025 | Sahiwal, Pakistan

- Architected, built, and shipped scalable cross-platform mobile applications for Android and iOS using Flutter and Dart, supporting thousands of active users.
- Engineered cloud backend integrations with Firebase services (Authentication, Cloud Firestore, Cloud Functions, and Cloud Storage), implementing offline sync, pagination, and state management.
- Managed staged release lifecycles across Google Play Store and Apple TestFlight; actively collaborated in Agile sprints, technical specifications, and peer code reviews.

Data Science & AI Bootcamp — DataCrumbs 05/2024 | Intensive ML Training

- Intensive training in modern Python data workflows: exploratory data analysis, feature engineering, cross-validation architectures, and production model evaluation.

EDUCATION

Bachelor of Science in Computer Science (BSCS) 10/2020 – 06/2024

Pir Mehr Ali Shah Arid Agriculture University (PMAS-AAUR) — Barani Institute of Sciences, Sahiwal, Pakistan

- **Final Grade / CGPA:** 3.28 / 4.00 | **Medium of Instruction:** 100% English (Official English Proficiency Certificate issued)
- **Thesis:** AI-Based Food Waste & Donation Management System (Frugal) — Flutter, Firebase, Python ML Extension
- **Relevant Coursework:** Artificial Intelligence, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Parallel & Distributed Computing

VERIFIED CERTIFICATIONS & ACADEMIC CREDENTIALS

Machine Learning Specialization — Stanford University & DeepLearning.AI (Coursera) Verified

- Supervised learning (linear/logistic regression, decision trees, neural networks), unsupervised learning (clustering, anomaly detection, recommender systems) by Andrew Ng.

Deep Learning Specialization (5-Course Specialization) — DeepLearning.AI (Coursera) Verified

- Neural networks, deep forward/backprop, hyperparameter tuning, regularization, optimization (Adam/RMSProp), CNNs, and sequence models (RNN/LSTM/GRU).

Mathematics for Machine Learning Specialization (3 Courses) — Imperial College London (Coursera) Verified

- Rigorous foundation in Linear Algebra (vector spaces, eigenvalues/vectors), Multivariate Calculus (gradients, Jacobians, Hessians), and PCA.

Google Data Analytics Professional Certificate (9 Courses) — Google (Coursera) Verified

- Comprehensive data preparation, statistical analysis, SQL querying, data cleaning, Tableau visualization, and analytical problem-solving.

Introduction to Machine Learning in Production (MLOps) — DeepLearning.AI (Coursera) Verified

- Production ML system design: scoping, data pipelines, monitoring concept and data drift, deployment strategies, and continuous model improvement.

Official English Proficiency Certificate — PMAS-AAUR Verified

- Certified by Registrar that the 4-year BS Computer Science curriculum, coursework, examinations, and final defense were delivered entirely in English.

RESEARCH INTERESTS & ENGINEERING STANDARDS

Core Focus Areas:	Trustworthy & Calibrated Machine Learning, Production MLOps & Continuous Delivery, Low-Resource Multilingual NLP, and Edge Computer Vision for Precision Agriculture.
Zero Data Leakage:	Strict time-aware rolling-origin backtests and entity-isolated splitting; zero target bleed across training and evaluation splits.
Uncertainty Honesty:	Prioritizes calibrated probability distributions (Platt scaling, ECE minimization, empirical prediction intervals) over raw overconfident logits.
Production Gates:	Automated schema validation (Great Expectations), distribution drift detection (KS & PSI), and input quality guards (Laplacian blur gates).

LANGUAGE PROFICIENCY & PERSONAL DETAILS

- **Urdu:** Native language (Full professional & spoken fluency).
- **English:** Professional Working Proficiency (CEFR B2 Level: Listening B2, Reading B2, Writing B2, Spoken Interaction B1+). All 4 years of university education completed, examined, and defended in 100% English.
- **Availability:** Available for immediate start; open to remote roles, hybrid setups, international relocation, and research-focused Master's opportunities.